UNIVERSITY OF THESSALY
SCHOOL OF ENGINEERING

DEPARTMENT OF ELECTRICAL
& COMPUTER ENGINEERING

Νευρο-Ασαφής Υπολογιστική (Neuro-Fuzzy Computing)
Fall Semester 2024-2025
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Problem-set 1st: **Teamwork** (2-persons)

Announcement: Monday, November 04, 2024
Delivery deadline: Sunday, December 01, 2024

SECTION 0: Warming-up with linear algebra, calculus, and automatic control



Problem-01

Plot the contour lines of the following function: $f(x, y) = 4 - \frac{1}{4}(x^2 + y^2)$. Write it in quadratic form. Then, find and characterize the (local) minima/maxima of this function (Show your analytic calculations).



Problem-02

Execute by hand two iterations of the Gradient Descent to the function $f(x_1, x_2)=0.26(x_1^2+x_2^2)-0.48(x_1x_2)$ with initial point $\mathbf{x_0}=(2, 3)$. Show your analytic calculations using the optimal λ in each iteration.



Problem-03

Consider the discrete dynamical system

$$x_{k+1} = f(x_k)$$
$$= c*x_k(1-x_k)$$

for x values in (0,1) and $0<c\leq 4$. Many dynamical systems transition into chaos as we increase a control or gain parameter such as c. Select c=3.5 and use the two choices of initial conditions, $x_0=0.5$ and $x_0=0.51$, to generate $x_1,\ldots,x_{20}$.

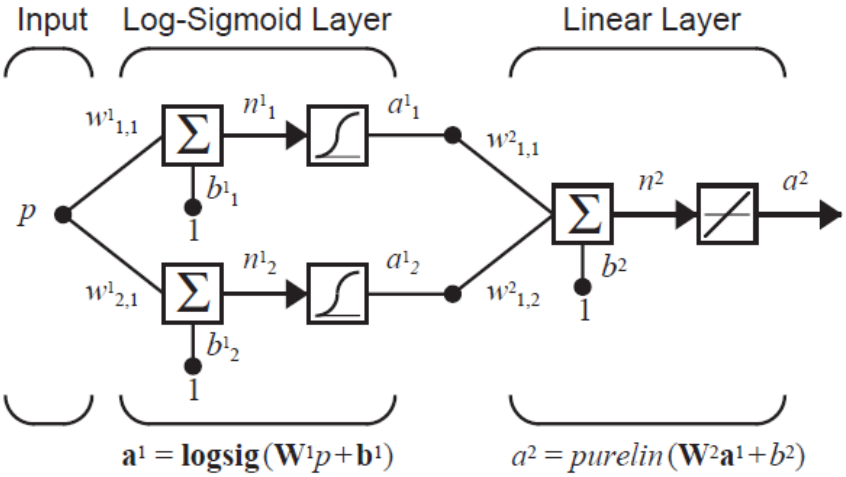
A. Plot the two trajectories. Are they aperiodic (chaotic) or periodic? Does a difference in 0.01 in initial condition significantly affect the overall shape of the discrete trajectory?
B. Now repeat the above experiment but use the gain parameter c=3.9 (or c=4). No matter how close two initial conditions, in a chaotic dynamical system they always produce divergent trajectories. Does c=3.9 produce chaos?

SECTION 1: Introduction to neural networks



Problem-04

Consider the following neural network



with $w^1_{1,1}$=-3, $w^1_{2,1}$=-1.5, b^1_1=-1, b^1_2=-0.25, $w^2_{1,1}$=1, $w^2_{1,2}$=1.5, b^2=2.
Sketch the following responses (plot the indicated variable versus p for -3 < p < 3).

A. a^1_1
B. a^1_2
C. a^2

Then, change the *logsig* activation function with the *ReLU* activation function and sketch again the aforementioned responses.



Problem-05

Derive the gradient descent rule when the error function is defined as follows:

$$E = \frac{1}{2}\sum_k (t_k - y_k)^2 + \sum_{i,j} w_{ji}^2$$

where y_k is the output of the network for the k-th input..



Problem-06

You are given the following activation function:

$$f(x) = \ln(1 + e^{A(x+B)+Cx^2}) - \ln(1 + e^{D(x-B)}) + E$$

where, A,B,C,D and E are trainable parameters.

A. Show (graphically and analytically) how this activation function can approximate, ReLU and sigmoid.

SECTION 2: Working with ADALINE neural networks



Problem-07

Suppose that we have the following four reference patterns and their targets:

$$\left\{\mathbf{p}_1 = \begin{bmatrix}-1\\2\end{bmatrix}, t_1 = -1\right\}\left\{\mathbf{p}_2 = \begin{bmatrix}2\\-1\end{bmatrix}, t_2 = -1\right\}\left\{\mathbf{p}_3 = \begin{bmatrix}0\\-1\end{bmatrix}, t_3 = 1\right\}\left\{\mathbf{p}_4 = \begin{bmatrix}-1\\0\end{bmatrix}, t_4 = 1\right\}$$

The probability of vector $\mathbf{p}_1$ is 0.25, the probability of vector $\mathbf{p}_2$ is 0.35, and the probability of vectors $\mathbf{p}_3$ and $\mathbf{p}_4$ is 0.20.

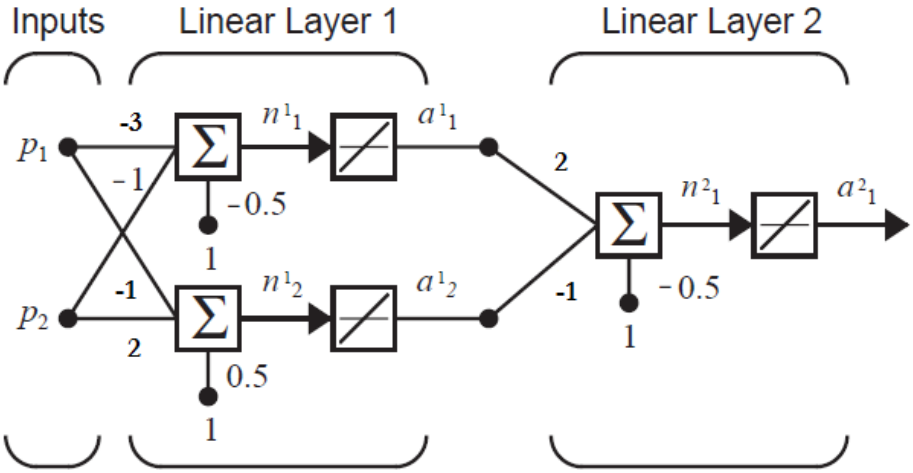
A. Draw the network diagram for a single-neuron ADALINE network with no bias that could be trained on these patterns.
B. Sketch the contour plot of the mean square error performance index.
C. Show the optimal decision boundary (for the weights that minimize mean square error), and verify (?) that it separates the patterns into the appropriate categories.
D. Find the maximum stable learning rate for the LMS algorithm.

E. Perform one iteration of the LMS algorithm, starting with all weights equal to zero, and presenting input vector $\mathbf{p}_1$. Use a learning rate of α=0.005.



Problem-08

Find a single-layer network that has the same input/output characteristic as the network below.





Problem-09

The patterns (0,0), (0,1), (1,0), (-1,-1) belong to class A and the patterns (2.1,0), (0,-2,5), (1.6,-1,6) belong to class B. The patterns of class A have probability equal to 0.10, whereas that of class B have probability each equal to 0.20.

A. Draw the patterns in a 2-D diagram. Are the categories linearly separable?
B. For the aforementioned patterns design the architecture of an ADALINE neural network which will separate the classes.
C. Starting with initial weights/biases equal to a random number in [-0.5, …, 0.5] calculate the final (after convergence) weights/biases of the ADALINE using the LMS training algorithm with learning rate=0.001.

SECTION 3: Basics of MLPs



Problem-10

Write code to implement the backpropagation algorithm for a 1-S^1-1 MLP network (logsigmoid-linear). Write the program using matrix operations, as we did in the class lecture. Choose the initial weights and biases to be random numbers uniformly distributed between -0.5 and 0.5, and train the network to approximate the function

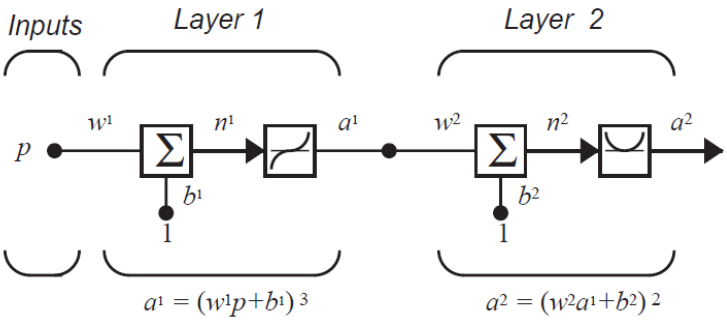
$$g(p)=1+\sin[p(\pi/3)] \text{ for } -2 \le p \le 2.$$

Use S^1= 2, S^1= 6 and S^1= 10. Experiment with several different values for the learning rate α, and use several different initial conditions. Discuss the convergence properties of the algorithm as the learning rate changes.



Problem-11

For the network shown below



the initial weights and biases are chosen to be

$w^1(0)= -1, b^1(0)= -0.5, \ w^2(0)= 2, b^2(0)= -1.$

An input/target pair is given to be {p=2, t=1}.

Perform one iteration of backpropagation with $\alpha=1$.

Useful information:

The deadline is strict. It is possible to get an extension (up to 4 days), but you need to get the approval of the instructor, and this is going to cost a 10% penalty to the final grade of this Problem-Set. Delivery of the solutions' pdf (typeset or very very easy to read handwritten) is done by email to dkatsar@uth.gr. The subject of the message should be **strictly**: CE418-Problem set 01: AEM1-AEM2

Symbol interpretation:



It requires "algorithmic" and/org mathematical thinking.



It requires the development of code (in any language/platform you wish). The final deliverable should contain: a) The solution of the exercise, and b) The implementation source code.